

MPLS L3VPN Inter-AS Option B

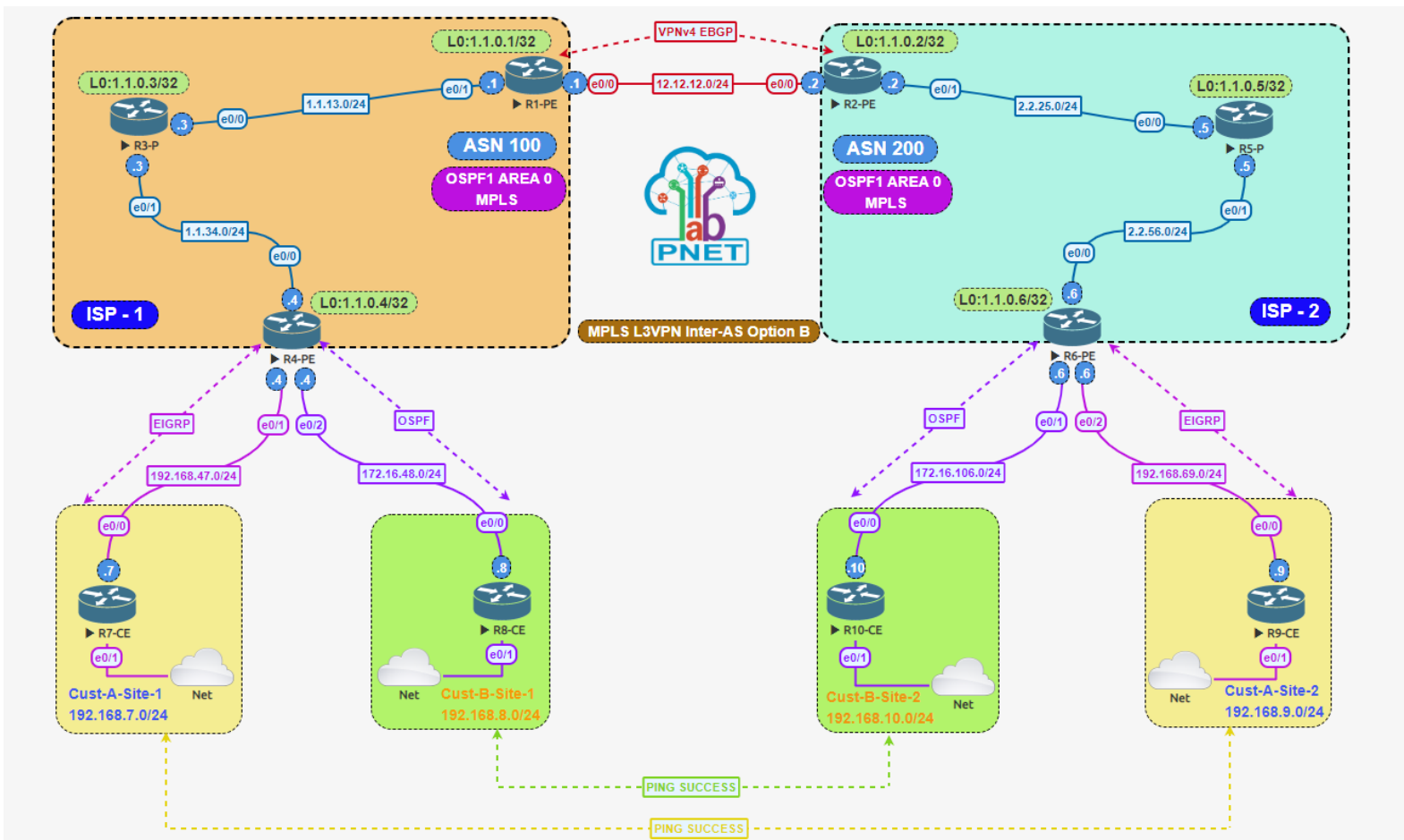
Lab Objective:

MPLS L3VPN Inter-AS Option B aka MPLS VPN Inter-AS with ASBRs Exchanging VPN-IPv4 Addresses, is similar to the previous Option A example with the exception that the Inter-AS link runs a single VPNv4 EBGP peering instead of multiple VRF aware IGP or BGP instances.

In this procedure, the PE routers use IBGP to redistribute labeled VPN-IPv4 routes either to an Autonomous System Border Router (ASBR), or to a route reflector of which an ASBR is a client. The ASBR then uses EBGP to redistribute those labeled VPN-IPv4 routes to an ASBR in another AS, which in turn distributes them to the PE routers in that AS, or perhaps to another ASBR which in turn distributes them, and so on.

Lab Topology:

The lab network topology is illustrated below:



Task 1: Basic Configuration.

Configure hostnames and IP addresses on all routers as illustrated in the network topology

Task 2: MPLS, IGP Core Routing.

Configure IGP routing and LDP in the first AS 100, which consists of R1, R3, and R4 as follows:

- Enable OSPF Area 0 on the links between R1 & R3 and R3 & R4.
- Enable OSPF Area 0 on the Loopback0 interfaces of R1, R3, & R4 as passive interfaces.
- Enable LDP on the links between R1 & R3 and R3 & R4.

Configure IGP routing and LDP in the first AS 102, which consists of R2, R5, and R6 as follows:

- Enable OSPF Area 0 on the links between R2 & R5 and R5 & R6.
- Enable OSPF Area 0 on the Loopback0 interfaces of R2, R5, & R6 as passive interfaces.
- Enable LDP on the links between R2 & R5 and R5 & R6.

Task 3: BGP VPNv4, Routing PE-CE.

Configure VPNv4 BGP peerings as follows:

- R1 and R4 should peer in AS 100 using each other's Loopback0 interfaces.
- R2 and R6 should peer in AS 200 using each other's Loopback0 interfaces.

Configure the following VRFs on PE routers R4 in AS 100 as follows:

- VRF Cust-A:
 - Route Distinguisher: 101:201
 - Route Target Import: 101:201
 - Route Target Export: 101:201
 - Assign this VRF on the links in the 192.168.47.0/24 network on R4 and R7.
- VRF Cust-B:
 - Route Distinguisher: 102:202
 - Route Target Import: 102:202
 - Route Target Export: 102:202
 - Assign this VRF on the links in the 172.16.48.0/24 network on R4 and R8.

Configure the following VRFs on PE routers R6 in AS 200 as follows:

- VRF Cust-A:
 - Route Distinguisher: 101:201
 - Route Target Import: 101:201
 - Route Target Export: 101:201



- Assign this VRF on the links in the 192.168.69.0/24 network on R6 and R9.
- VRF Cust-B:
 - Route Distinguisher: 102:202
 - Route Target Import: 102:202
 - Route Target Export: 102:202
 - Assign this VRF on the links in the 172.16.106.0/24 network on R6 and R10.

Configure EIRGP routing for VRF Cust-A as follows:

- Enable EIRGP between R4 & R7.
- Enable EIRGP between R6 & R9.
- Advertise the Ethernet networks of R7 & R9 into EIRGP.

Configure OSPF routing for VRF Cust-B as follows:

- Use OSPF 100 Area 0
- Enable OSPF between R4 & R8.
- Enable OSPF between R6 & R10.
- Advertise Ethernet networks of R8 & R10 into OSPF.

Redistribute between VPNv4 BGP and the VRF aware IGP processes on the PE routers R4, R6.

Task 4: MPLS L3VPN Inter-AS Option B between R4 and R6

Configure MPLS L3VPN Inter-AS Option B between R4 and R6

Once complete the following reachability should be achieved:

- Customer routers R7 and R9 should have full IP reachability to each other's networks.
- Customer routers R8 and R10 should have full IP reachability to each other's networks.

Solution

Task 1: Basic Configuration

Configure hostnames and IP addresses on all routers as illustrated in the network topology

R1

hostname R1-PE

R2

hostname R2-PE



R3

hostname R3-P

R4

hostname R4-PE

R5

hostname R5-P

R6

hostname R6-PE

R7

hostname R7-CE

R8

hostname R8-CE

R9

hostname R9-CE

R10

hostname R10-CE

Task 2: MPLS, IGP Core Routing

Configure IGP routing and LDP in the first AS 100, which consists of R1, R3, and R4 as follows:

- Enable OSPF Area 0 on the links between R1 & R3 and R3 & R4.
- Enable OSPF Area 0 on the Loopback0 interfaces of R1, R3, & R4 as passive interfaces.
- Enable LDP on the links between R1 & R3 and R3 & R4.

Configure IGP routing and LDP in the first AS 102, which consists of R2, R5, and R6 as follows:

- Enable OSPF Area 0 on the links between R2 & R5 and R5 & R6.



- Enable OSPF Area 0 on the Loopback0 interfaces of R2, R5, & R6 as passive interfaces.
- Enable LDP on the links between R2 & R5 and R5 & R6.

R1 (PE Router) – ISP1 - ASBR connected to ISP 2

```
interface Loopback0
 ip address 1.1.0.1 255.255.255.255
 ip ospf 1 area 0
!
interface Ethernet0/1
 no shut
 ip address 1.1.13.1 255.255.255.0
 ip ospf network point-to-point
 ip ospf 1 area 0
 mpls ip
!
interface Ethernet0/0
 no shut
 ip address 12.12.12.1 255.255.255.0
!
router ospf 1
 router-id 1.1.0.1
```

R3 (P Router) - ISP 1

```
interface Loopback0
 ip address 1.1.0.3 255.255.255.255
 ip ospf 1 area 0
!
interface Ethernet0/0
 no shut
 ip address 1.1.13.3 255.255.255.0
 ip ospf network point-to-point
 ip ospf 1 area 0
 mpls ip
!
interface Ethernet0/1
 no shut
 ip address 1.1.34.3 255.255.255.0
 ip ospf network point-to-point
 ip ospf 1 area 0
 mpls ip
!
router ospf 1
```





```
router-id 1.1.0.3
```

R4 (PE Router) - ISP 1

```
interface Loopback0
 ip address 1.1.0.4 255.255.255.255
 ip ospf 1 area 0
!
interface Ethernet0/0
 no shut
 ip address 1.1.34.4 255.255.255.0
 ip ospf network point-to-point
 ip ospf 1 area 0
 mpls ip
!
router ospf 1
 router-id 1.1.0.4
```

R2 (PE Router) - ISP 2 - ASBR connected to ISP 1

```
interface Loopback0
 ip address 2.2.0.2 255.255.255.255
 ip ospf 1 area 0
!
interface Ethernet0/1
 no shut
 ip address 2.2.25.2 255.255.255.0
 ip ospf network point-to-point
 ip ospf 1 area 0
 mpls ip
!
interface Ethernet0/0
 no shut
 ip address 12.12.12.2 255.255.255.0

router ospf 1
 router-id 2.2.0.2
```

R5 (P Router) - ISP 2

```
interface Loopback0
 ip address 2.2.0.5 255.255.255.255
 ip ospf 1 area 0
!
interface Ethernet0/0
 no shut
 ip address 2.2.25.5 255.255.255.0
```





```
ip ospf network point-to-point
ip ospf 1 area 0
mpls ip
!
interface Ethernet0/1
no shut
ip address 2.2.56.5 255.255.255.0
ip ospf network point-to-point
ip ospf 1 area 0
mpls ip
!
router ospf 1
router-id 2.2.0.5
```

R6 (PE Router) - ISP 2

```
interface Loopback0
ip address 2.2.0.6 255.255.255.255
ip ospf 1 area 0
!
interface Ethernet0/0
no shut
ip address 2.2.56.6 255.255.255.0
ip ospf network point-to-point
ip ospf 1 area 0
mpls ip
!
router ospf 1
router-id 2.2.0.6
```

Task 3: BGP VPNv4, Routing PE-CE.

Configure VPNv4 BGP peerings as follows:

- R1 and R4 should peer in AS 100 using each other's Loopback0 interfaces.
- R2 and R6 should peer in AS 200 using each other's Loopback0 interfaces.

Configure the following VRFs on PE routers R4 in AS 100 as follows:

- VRF Cust-A:
 - Route Distinguisher: 101:201
 - Route Target Import: 101:201
 - Route Target Export: 101:201
 - Assign this VRF on the links in the 192.168.47.0/24 network on R4 and R7.
- VRF Cust-B:





- Route Distinguisher: 102:202
- Route Target Import: 102:202
- Route Target Export: 102:202
- Assign this VRF on the links in the 172.16.48.0/24 network on R4 and R8.

Configure the following VRFs on PE routers R6 in AS 200 as follows:

- VRF Cust-A:
 - Route Distinguisher: 101:201
 - Route Target Import: 101:201
 - Route Target Export: 101:201
 - Assign this VRF on the links in the 192.168.69.0/24 network on R6 and R9.
- VRF Cust-B:
 - Route Distinguisher: 102:202
 - Route Target Import: 102:202
 - Route Target Export: 102:202
 - Assign this VRF on the links in the 172.16.106.0/24 network on R6 and R10.

Configure EIRGP routing for VRF Cust-A as follows:

- Enable EIRGP between R4 & R7.
- Enable EIRGP between R6 & R9.
- Advertise the Ethernet networks of R7 & R9 into EIRGP.

Configure OSPF routing for VRF Cust-B as follows:

- Use OSPF 100 Area 0
- Enable OSPF between R4 & R8.
- Enable OSPF between R6 & R10.
- Advertise Ethernet networks of R8 & R10 into OSPF.

Redistribute between VPNv4 BGP and the VRF aware IGP processes on the PE routers R4, R6.

R1:

```
router bgp 100
  bgp router-id 1.1.0.1
  bgp log-neighbor-changes
  no bgp default ipv4-unicast
  no bgp default route-target filter
  neighbor 1.1.0.4 remote-as 100
  neighbor 1.1.0.4 update-source Loopback0
  !
  address-family vpnv4
```





```
neighbor 1.1.0.4 activate
neighbor 1.1.0.4 send-community extended
neighbor 1.1.0.4 next-hop-self
```

R4:

```
router bgp 100
  bgp router-id 1.1.0.4
  bgp log-neighbor-changes
  no bgp default ipv4-unicast
  neighbor 1.1.0.1 remote-as 100
  neighbor 1.1.0.1 update-source Loopback0
  !
  address-family ipv4
  exit-address-family
  !
  address-family vpnv4
    neighbor 1.1.0.1 activate
    neighbor 1.1.0.1 send-community extended
  exit-address-family
```

R2:

```
router bgp 200
  bgp router-id 2.2.0.2
  bgp log-neighbor-changes
  no bgp default ipv4-unicast
  no bgp default route-target filter
  neighbor 2.2.0.6 remote-as 200
  neighbor 2.2.0.6 update-source Loopback0
  !
  address-family vpnv4
    neighbor 2.2.0.6 activate
    neighbor 2.2.0.6 send-community extended
    neighbor 2.2.0.6 next-hop-self
  exit-address-family
```

R6:

```
router bgp 200
  bgp router-id 2.2.0.6
  bgp log-neighbor-changes
  no bgp default ipv4-unicast
  neighbor 2.2.0.2 remote-as 200
```





```
neighbor 2.2.0.2 update-source Loopback0
!  
address-family ipv4  
exit-address-family  
!  
address-family vpnv4  
neighbor 2.2.0.2 activate  
neighbor 2.2.0.2 send-community extended  
exit-address-family
```

R4:

```
vrf definition Cust-A  
rd 101:201  
!  
address-family ipv4  
route-target export 101:201  
route-target import 101:201  
exit-address-family  
!  
vrf definition Cust-B  
rd 102:202  
!  
address-family ipv4  
route-target export 102:202  
route-target import 102:202  
exit-address-family  
!  
interface Ethernet0/1  
no shut  
vrf forwarding Cust-A  
ip address 192.168.47.4 255.255.255.0  
!  
interface Ethernet0/2  
no shut  
vrf forwarding Cust-B  
ip address 172.16.48.4 255.255.255.0  
ip ospf network point-to-point  
ip ospf 100 area 0  
  
router eigrp 1  
!  
address-family ipv4 vrf Cust-A autonomous-system 100  
default-metric 10000 100 255 1 1500  
redistribute bgp 100
```





```
network 192.168.47.0
exit-address-family
!
router ospf 100 vrf Cust-B
router-id 172.16.48.4
redistribute bgp 100 subnets
!
router bgp 100
bgp router-id 1.1.0.4
address-family ipv4 vrf Cust-A
redistribute connected
redistribute eigrp 100
exit-address-family
!
address-family ipv4 vrf Cust-B
redistribute connected
redistribute ospf 100 match internal external 1 external 2
exit-address-family
```

R6:

```
vrf definition Cust-A
rd 101:201
!
address-family ipv4
route-target export 101:201
route-target import 101:201
exit-address-family
!
vrf definition Cust-B
rd 102:202
!
address-family ipv4
route-target export 102:202
route-target import 102:202
exit-address-family
!
!
interface Ethernet0/1
no shut
vrf forwarding Cust-B
ip address 172.16.106.6 255.255.255.0
ip ospf network point-to-point
ip ospf 200 area 0
!
```





```
interface Ethernet0/2
  no shut
  vrf forwarding Cust-A
  ip address 192.168.69.6 255.255.255.0
!
router eigrp 1
!
  address-family ipv4 vrf Cust-A autonomous-system 100
    default-metric 10000 100 255 1 1500
    redistribute bgp 200
    network 192.168.69.0
  exit-address-family
!
router ospf 200 vrf Cust-B
  router-id 172.16.106.6
  redistribute bgp 200 subnets
!
router bgp 200
  bgp router-id 2.2.0.6
  address-family ipv4 vrf Cust-A
    redistribute connected
    redistribute eigrp 100
  exit-address-family
!
  address-family ipv4 vrf Cust-B
    redistribute connected
    redistribute ospf 200 match internal external 1 external 2
  exit-address-family
```

R7 (CE Router) - Customer A Site 1

EIGRP is the PE-CE routing protocol used for this customer

```
interface Ethernet0/0
  no shut
  ip address 192.168.47.7 255.255.255.0
!
interface Ethernet0/1
  no shut
  ip address 192.168.7.7 255.255.255.0
!
router eigrp 100
  network 192.168.0.0 0.0.255.255
```

R8 (CE Router) - Customer B Site 1





OSPF is the PE-CE routing protocol used for this customer

```
interface Ethernet0/0
  no shut
  ip address 172.16.48.8 255.255.255.0
  ip ospf network point-to-point
  ip ospf 1 area 0
!
interface Ethernet0/1
  no shut
  ip address 172.16.8.8 255.255.255.0
  ip ospf network point-to-point
  ip ospf 1 area 0
!
router ospf 1
  router-id 172.16.8.8
```

R9 (CE Router) - Customer A Site 2

EIGRP is the PE-CE routing protocol used for this customer

```
interface Ethernet0/0
  no shut
  ip address 192.168.69.9 255.255.255.0
!
interface Ethernet0/1
  no shut
  ip address 192.168.9.9 255.255.255.0
!
router eigrp 100
  network 192.168.0.0 0.0.255.255
```

R10 (CE Router) - Customer B Site 2

OSPF is the PE-CE routing protocol used for this customer

```
interface Ethernet0/0
  no shut
  ip address 172.16.106.10 255.255.255.0
  ip ospf network point-to-point
  ip ospf 1 area 0
!
interface Ethernet0/1
  no shut
  ip address 172.16.10.10 255.255.255.0
```





```
ip ospf network point-to-point
ip ospf 1 area 0
!
router ospf 1
router-id 172.16.10.10
```

Task 4: MPLS L3VPN Inter-AS Option B between R4 and R6

Configure MPLS L3VPN Inter-AS Option B between R4 and R6

Once complete the following reachability should be achieved:

- Customer routers R7 and R9 should have full IP reachability to each other's networks.
- Customer routers R8 and R10 should have full IP reachability to each other's networks.

R1:

```
interface Ethernet0/0
no shut
ip address 12.12.12.1 255.255.255.0
mpls bgp forwarding
!
router bgp 100
neighbor 12.12.12.2 remote-as 200
!
address-family vpnv4
neighbor 12.12.12.2 activate
neighbor 12.12.12.2 send-community extended
exit-address-family
```

R2:

```
interface Ethernet0/0
no shut
ip address 12.12.12.2 255.255.255.0
mpls bgp forwarding
!
router bgp 200
neighbor 12.12.12.1 remote-as 100
!
address-family vpnv4
neighbor 12.12.12.1 activate
neighbor 12.12.12.1 send-community extended
exit-address-family
```





To verify the configuration:

R7-CE#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS

level-2

ia - IS-IS inter area, * - candidate default, U - per-user

static route

o - ODR, P - periodic downloaded static route, H - NHRP, l -

LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.7.0/24 is directly connected, Ethernet0/1

L 192.168.7.7/32 is directly connected, Ethernet0/1

D 192.168.9.0/24 [90/332800] via 192.168.47.4, 00:01:46,
Ethernet0/0

192.168.47.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.47.0/24 is directly connected, Ethernet0/0

L 192.168.47.7/32 is directly connected, Ethernet0/0

D 192.168.69.0/24 [90/307200] via 192.168.47.4, 00:01:46,
Ethernet0/0

R8-CE#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS

level-2

ia - IS-IS inter area, * - candidate default, U - per-user

static route

o - ODR, P - periodic downloaded static route, H - NHRP, l -

LISP

a - application route



+ - replicated route, % - next hop override

Gateway of last resort is not set

```
172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C       172.16.8.0/24 is directly connected, Ethernet0/1
L       172.16.8.8/32 is directly connected, Ethernet0/1
O E2    172.16.10.0/24 [110/1] via 172.16.48.4, 00:08:34, Ethernet0/0
C       172.16.48.0/24 is directly connected, Ethernet0/0
L       172.16.48.8/32 is directly connected, Ethernet0/0
O E2    172.16.106.0/24 [110/1] via 172.16.48.4, 00:08:34, Ethernet0/0
```

Ping Customer-A Site 2 network

```
R7-CE#ping 192.168.9.9 source 192.168.7.7
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.9.9, timeout is 2 seconds:

Packet sent with a source address of 192.168.7.7

!!!!

Traceroute Customer-A Site 2 network

```
R7-CE#traceroute 192.168.9.9 source 192.168.7.7
```

Type escape sequence to abort.

Tracing the route to 192.168.9.9

VRF info: (vrf in name/id, vrf out name/id)

```
 1 192.168.47.4 2 msec 2 msec 2 msec
 2 1.1.34.3 [MPLS: Labels 17/24 Exp 0] 9 msec 10 msec 10 msec
 3 1.1.13.1 [MPLS: Label 24 Exp 0] 6 msec 15 msec 14 msec
 4 12.12.12.2 [MPLS: Label 19 Exp 0] 13 msec 8 msec 6 msec
 5 2.2.25.5 [MPLS: Labels 17/19 Exp 0] 7 msec 9 msec 8 msec
 6 192.168.69.6 [MPLS: Label 19 Exp 0] 6 msec 12 msec 6 msec
 7 192.168.69.9 8 msec 22 msec *
```

We can see from the above output there are three different LSP (Label Switch Path) are being used to provide end-to-end reachability.

LSP-1: From R4 to R1 using VPN Label 24

```
R4-PE#sh bgp vpnv4 unicast vrf Cust-A 192.168.9.0
```

*Aug 1 15:22:44.995: %SYS-5-CONFIG_I: Configured from console by console

```
R4-PE#sh bgp vpnv4 unicast vrf Cust-A 192.168.9.0
```

BGP routing table entry for 101:201:192.168.9.0/24, version 10

Paths: (1 available, best #1, table Cust-A)

Not advertised to any peer

Refresh Epoch 2

200

1.1.0.1 (metric 21) from 1.1.0.1 (1.1.0.1)

Origin incomplete, metric 0, localpref 100, valid, internal, best

Extended Community: RT:101:201 0x8800:32768:0 0x8801:100:51200

0x8802:65281:256000 0x8803:65281:1500 0x8806:0:3232253193

mpls labels in/out nolabel/24

rx pathid: 0, tx pathid: 0x0

LSP-2: From R1 to R2 using VPN Label 19

R1-PE#sh bgp vpnv4 unicast all 192.168.9.0

BGP routing table entry for 101:201:192.168.9.0/24, version 6

Paths: (1 available, best #1, no table)

Advertised to update-groups:

1

Refresh Epoch 1

200

12.12.12.2 from 12.12.12.2 (2.2.0.2)

Origin incomplete, localpref 100, valid, external, best

Extended Community: RT:101:201 0x8800:32768:0 0x8801:100:51200

0x8802:65281:256000 0x8803:65281:1500 0x8806:0:3232253193

mpls labels in/out 24/19

rx pathid: 0, tx pathid: 0x0

LSP-3: From R2 to R6 using VPN label 19

R2-PE#sh bgp vpnv4 unicast all 192.168.9.0

BGP routing table entry for 101:201:192.168.9.0/24, version 2

Paths: (1 available, best #1, no table)

Advertised to update-groups:

2

Refresh Epoch 1

Local

2.2.0.6 (metric 21) from 2.2.0.6 (2.2.0.6)

Origin incomplete, metric 307200, localpref 100, valid, internal,
best

Extended Community: RT:101:201 Cost:pre-bestpath:128:307200

0x8800:32768:0 0x8801:100:51200 0x8802:65281:256000

0x8803:65281:1500

0x8806:0:3232253193

mpls labels in/out 19/19

rx pathid: 0, tx pathid: 0x0

Finally unlabelled traffic from ISP 2 PE Router (R6) to CE Router (R9)



```
R6-PE#sh bgp vpnv4 unicast vrf Cust-A 192.168.9.0
```

```
*Aug 1 15:25:20.398: %SYS-5-CONFIG_I: Configured from console by console
```

```
R6-PE#sh bgp vpnv4 unicast vrf Cust-A 192.168.9.0
```

```
BGP routing table entry for 101:201:192.168.9.0/24, version 2
```

```
Paths: (1 available, best #1, table Cust-A)
```

```
Advertised to update-groups:
```

```
1
```

```
Refresh Epoch 1
```

```
Local
```

```
192.168.69.9 from 0.0.0.0 (2.2.0.6)
```

```
Origin incomplete, metric 307200, localpref 100, weight 32768,  
valid, sourced, best
```

```
Extended Community: RT:101:201 Cost:pre-bestpath:128:307200
```

```
0x8800:32768:0 0x8801:100:51200 0x8802:65281:256000
```

```
0x8803:65281:1500
```

```
0x8806:0:3232253193
```

```
mpls labels in/out 19/nolabel
```

```
rx pathid: 0, tx pathid: 0x0
```

